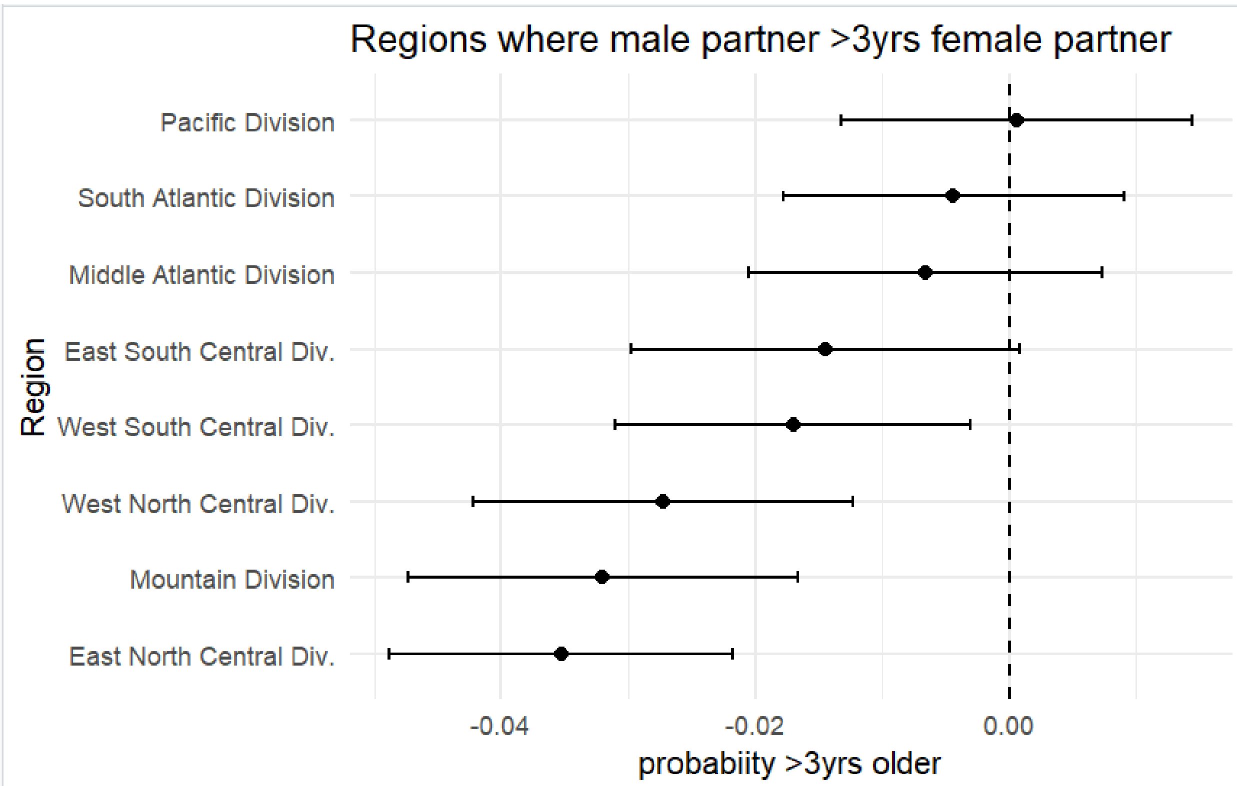
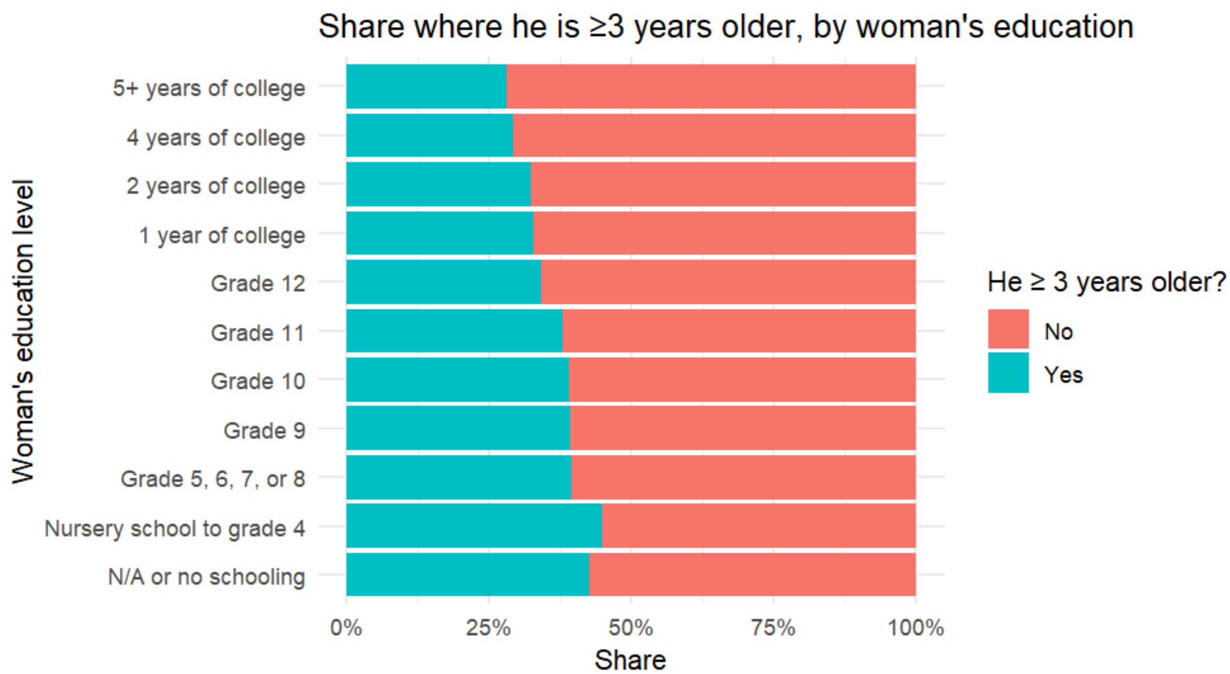
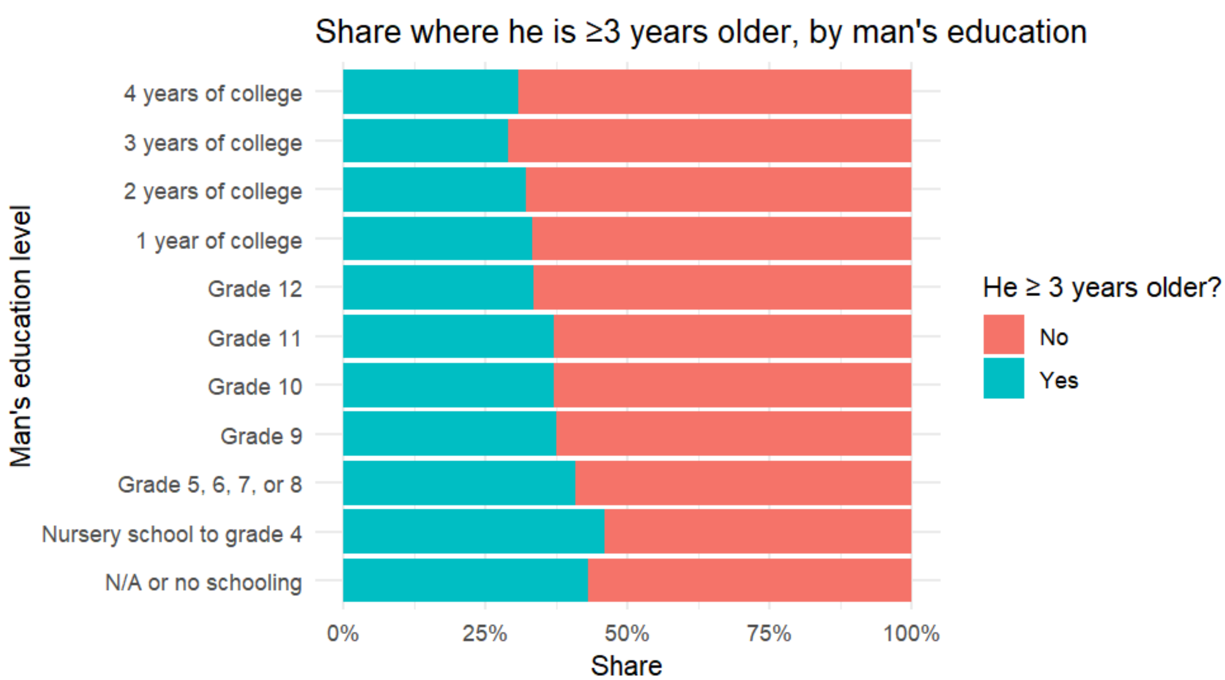
Aqeel Choudhury
11/6/25
B2000
HW 7



The coefficient plot graph shows a distribution of the probability of the male partner being 3 years or older than the female partner by region. I decided to look around the Northeast region. Surprisingly we see the only non negative region being the pacific division. The most negative division being East North Central whc\ich I did expect. I definitely expected the south atlantic division to be the most positive.





Next I compared the probability of the male partner being 3+ years older than the female partner by education level of the male partner and female partner. As we can see it is mostly similar but some slight differences between the probabilities. A lower probability the male partner is 3+ years older when the female partner has 4+ years of college experience than when he does and slightly higher probabilities when she has education levels grades 9-12 than when he does.

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library(ggplot2)
library(tidyverse)
library(haven)
library(tidyr)


load("ACS_2021_couples.RData")



acs2021_couples$RACE <- fct_recode(as.factor(acs2021_couples$RACE),
                 "White" = "1",
                 "Black" = "2",
                 "American Indian or Alaska Native" = "3",
                 "Chinese" = "4",
                 "Japanese" = "5",
                 "Other Asian or Pacific Islander" = "6",
                 "Other race" = "7",
                 "two races" = "8",
                 "three races" = "9")

acs2021_couples$h_race <- fct_recode(as.factor(acs2021_couples$h_race),
                  "White" = "1",
                  "Black" = "2",
                  "American Indian or Alaska Native" = "3",
                  "Chinese" = "4",
                  "Japanese" = "5",
                  "Other Asian or Pacific Islander" = "6",
                  "Other race" = "7",
                  "two races" = "8",
                  "three races" = "9")

acs2021_couples$HISPAN <- fct_recode(as.factor(acs2021_couples$HISPAN),
                  "Not Hispanic" = "0",
                  "Mexican" = "1",
                  "Puerto Rican" = "2",
                  "Cuban" = "3",
                  "Other" = "4")
acs2021_couples$h_hispan <- fct_recode(as.factor(acs2021_couples$h_hispan),
                   "Not Hispanic" = "0",
                   "Mexican" = "1",
                   "Puerto Rican" = "2",
                   "Cuban" = "3",
                   "Other" = "4")
trad_data <- acs2021_couples %>% filter( (SEX == "Female") & (h_sex == "Male") )

trad_data$he_more_than_3yrs_than_her <- as.numeric(trad_data$age_diff < -3)


table(trad_data$he_more_than_3yrs_than_her, cut(trad_data$age_diff, c(-100, -10, -3, 0, 3, 10, 100)))

trad_data <- trad_data %>%
  mutate(educ_numeric = case_when(
    educ_advdeg    == 1 ~ 18, educ_college   == 1 ~ 16, educ_somecoll  == 1 ~ 14, educ_hs        == 1 ~ 12, educ_nohs      == 1 ~ 10,
    TRUE ~ NA_real_ ))

trad_data <- trad_data %>%
  mutate(
   h_educ_chr = as.character(h_educ),
   h_educ_numeric = case_when(
    grepl("Graduate|Master|Professional|Doctor", h_educ_chr, ignore.case = TRUE) ~ 18,
    grepl("Bachelor", h_educ_chr, ignore.case = TRUE) ~ 16,
    grepl("Some college|Associate", h_educ_chr, ignore.case = TRUE) ~ 14,
    grepl("High school", h_educ_chr, ignore.case = TRUE) ~ 12,
    grepl("Less than|No schooling|Grade", h_educ_chr, ignore.case = TRUE) ~ 10,
```

```
    TRUE ~ NA_real_))

ols_out1 <- lm(he_more_than_3yrs_than_her ~ INCWAGE + educ_college + educ_advdeg + AGE, data = trad_data)
summary(ols_out1)

ols_out2 <- lm(he_more_than_3yrs_than_her ~ educ_numeric + h_educ_numeric + AGE, data = trad_data)
summary(ols_out2)

ols_out3 <- lm(he_more_than_3yrs_than_her ~ EDUC + h_educ + AGE, data = trad_data)
summary(ols_out3)

trad_data <- trad_data %>%
  mutate(REGION = fct_relevel(as.factor(REGION), "Northeast"))

ols_out5 <- lm(he_more_than_3yrs_than_her ~ educ_numeric + h_educ_numeric + AGE + REGION, data = trad_data,
na.action = na.omit)
summary(ols_out5)

region_coefs <- tidy(ols_out5, conf.int = TRUE) %>%
  filter(grepl("^REGION", term)) %>%
  mutate(region = sub("^REGION", "", term))

ggplot(region_coefs, aes(x = reorder(region, estimate), y = estimate)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0.15) +
  coord_flip() +
  labs(
    x = "Region",
    y = "probabiity >3yrs older",
    title = "Regions where male partner >3yrs female partner",) +
  theme_minimal()



ggplot(trad_data, aes(x = as.factor(EDUC),
                      fill = factor(he_more_than_3yrs_than_her, labels = c("No", "Yes")))) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(x = "Woman's education level", y = "Share", fill = "He ≥ 3 years older?",
       title = "Share where he is ≥3 years older, by woman's education") +
  theme_minimal() +
  coord_flip()


ggplot(trad_data, aes(x = as.factor(h_educ),
                      fill = factor(he_more_than_3yrs_than_her, labels = c("No", "Yes")))) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(x = "Man's education level", y = "Share", fill = "He ≥ 3 years older?",
       title = "Share where he is ≥3 years older, by man's education") +
  theme_minimal() +
  coord_flip()
```